AMNA FIAZ

+92 321 9976050 | amna.fiaz.ai.ds@gmail.com | https://github.com/Amnafiaz200 | https://www.linkedin.com/in/amna-ai-engineer/ | Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer building production AI systems, agentic RAG assistants, document-verification platforms, and data pipelines on GCP and AWS. Cut end-to-end data-processing time by 40 per cent across 8 clinical datasets, and built collection pipelines handling 50,000+ records per run. Redis Associate Developer certified. Worked on projects involving LLMs, RAG, OCR, PDF verification, web automation, data engineering, and multi-agent workflows, with a strong focus on building reliable systems that solve real business problems.

TECHNICAL SKILLS

AI & LLM Systems	RAG Systems, LangGraph, LLM Agents, Tool Calling, FAISS, ChromaDB, Groq, HuggingFace, AWS Bedrock, Strands Harness, Prompt Engineering, LLM Evaluation
Techstack	Python, FastAPI, Flask, Django, REST APIs, GraphQL, Pydantic, SQLAlchemy, Uvicorn, async/await, Celery, React, Next.js, Vue.js
Data & Databases	PostgreSQL, MySQL, MongoDB, Pandas, NumPy, schema design, query optimization, data validation, Precision/Recall/F1
Document AI & Automation	PyMuPDF, PaddleOCR, LayoutLmv3, Playwright, PDF Parsing, Regex Extraction, n8n
Cloud & DevOps	Docker, Docker Compose, Git, GitHub Actions, Linux, AWS EC2, GCP, Cloud Composer, Redis, SSH

WORK EXPERIENCE

Black Lion Innovations Group — Data Scientist — Mar 2026 – Present

- Built a women's hormonal-health RAG assistant with agent-based workflows, trusted knowledge retrieval, conversation context, and personalized responses.
- Built a Spotify AI Data Intelligence Agent using AWS Strands Harness and MCP, enabling natural-language analysis of artists, playlists, trends, and performance data with context management and memory.
- Automated annotation for a gas-leakage detection dataset using the Segment Anything Model (SAM), then manually reviewed and adjusted the generated masks to improve annotation accuracy and dataset quality.

LUMS BIRL — AI Engineer — May 2025 – Feb 2026

- Implemented Python ETL pipelines and FastAPI services for cleaning, transforming, combining, and reporting data from 8 clinical datasets.
- Integrated dbt Fusion into data-processing workflows, reducing end-to-end processing time by around 40
- Designed SQL, Pandas, and NumPy workflows with dbt Fusion for feature engineering, data validation, and state-aware data transformations.

Independent Contractor — LUMS (Prof. Abdul Basit) — Automation Engineer — Project-based

- Built an automated document-verification platform for Spanish energy-efficiency audit claims, covering 11 document types and roughly 40–50 fields per document. Developed PDF parsing, OCR, field extraction, validation, and cross-document checks to compare names, IDs, addresses, prices, measurements, signatures, and audit data.
- Developed a transformer-based LayoutLMv3 classifier that uses text, layout, and annotation information to classify structured documents.

SKYCODE — Database Developer Intern — May 2024 – Aug 2024

- Designed and optimized MySQL and PostgreSQL databases, improving data retrieval speed by approximately 20%.
- Supported database backup, recovery, performance monitoring, and routine database maintenance for application workloads.
- Assisted with backend integration and SQL query optimization for database-driven web applications.

SELECTED PROJECTS

Automated Document Verification Platform – Spanish Energy-Efficiency Audits

- Built a document-verification platform for Spanish energy-efficiency audits, processing 11 document types and validating 40–50 fields including IDs, addresses, measurements, prices, signatures, and audit data.
- Combined PyMuPDF, PaddleOCR, PostgreSQL, and rule-based validation to extract, cross-check, and annotate PDF fields and manually reviewable information.

Coder Vista – AI-Powered Career Intelligence Platform

- Built a multi-agent AI career platform using Lang graph, OpenAI Agents SDK, and MCP for CV analysis, job discovery, and skill-based role matching.
- Developed personalized skill-gap analysis and learning roadmaps using job requirements, market demand, and automated background workflows.

ELSA – Women's Hormonal Health Agentic RAG Platform

- Built a personalized AI health assistant using RAG with AWS Bedrock, LangGraph, and ChromaDB to retrieve relevant medical knowledge and maintain conversation context.
- Added structured guidance, and privacy-aware workflows for repeated conversations across PMS, PCOS, mood, sleep, and thyroid-related topics.

Spotify AI Data Intelligence Agent

- Built a Spotify AI Data Intelligence Agent using AWS Strands Harness, with an MCP server exposing Spotify analytics tools and data sources for natural-language analysis of artists, playlists, trends, and performance data.
- Implemented context management, memory, result offloading, context compaction, and selective retention to make long-running analysis more efficient and reduce unnecessary processing, latency, and AI usage.

WhatsApp Visa Processing AI Agent

- Built an automated visa-processing workflow using n8n and Twilio WhatsApp API, allowing users to submit documents and information through WhatsApp.
- Integrated LlamaParse, Groq, Google Sheets, webhooks, and human approval steps to extract structured information and route cases for review.

CERTIFICATIONS

- **Redis Associate Developer** — Redis | Aug 2026 – Aug 2028 | View Credential
- **Agentic Systems with LangGraph** — DataCamp | 2026 | View Credential
- **Certificate: 3rd International AI Championship** — AIEF/LUMS | Nov 2025 | View Certificate
- **Amazon Elastic Container Service (AWS ECS)** — KodeKloud | Aug 2026 | View Credential
- **Model Context Protocol: Advanced Topics** — Anthropic | Aug 2026 | View Credential

EDUCATION

UET Lahore — BSc Computer Engineering — Sep 2021 – Mar 2025
CGPA: 3.58/4.0